Shuffle Vision Transformer: Lightweight, Fast and Efficient Recognition of Driver's Facial Expression (v3)

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1.Introduction

Driver Facial Expression Recognition (DFER) plays an important role in monitoring driver alertness and preventing accidents. The given paper proposes ShuffViT-DFER, a lightweight hybrid model that combines ShuffleNetV2 (CNN) for efficient local feature extraction and EfficientViT-M2 (Vision Transformer) for global contextual understanding. By fusing these complementary features, the model achieves strong recognition accuracy on the KMU-FED dataset while keeping computational cost low enough for realtime, in-car systems. This makes ShuffViT-DFER a practical solution for intelligent driver-assistance and safety-monitoring applications.

2. Research Objective

- Develop an accurate, efficient, and real-time DFER system.
- System should be robust enough to withstand the challenges exist on the road environment.
- Does redesigning the classifier head (MLP vs. Conv-GAP) improve the model's ability to fuse CNN and ViT features for emotion recognition?
- **Improve road safety by mitigating the potential risks caused by human errors.**

Fig. 1: Driving environment challenges

3. Proposed Methodology

1) Preprocessing & Feature Extraction: We utilize the KMU-FED dataset. Faces are detected (MTCNN) and resized to 224x224, applying data augmentation. The model employs transfer learning to simultaneously extract features by fusing the outputs of two lightweight models: **ShuffleNet V2** (CNN) and **EfficientViT-M2** (ViT).

2) Classifier Head Extensions: After feature fusion, the original single fully connected (FC) classifier will **re-designed** to test aggregation strategies:

- **MLP Head:** two FC layers with batch-norm, ReLU, and dropout.
- **Conv-GAP Head:** 1×1 convolution + Global Average Pooling + FC.

3) Training & Evaluation: Models are trained for **90 epochs (25 epochs) and 10 folds (3 folds)** using the **Adam optimizer** (LR: 0.001, Batch Size: 128) on KMU-FED. Performance is comprehensively evaluated using **Accuracy, Precision, Recall, and F1-score**, alongside efficiency metrics like **Parameter Count** and **Processing Time**, and interpretability via visualization (Grad-Cam).

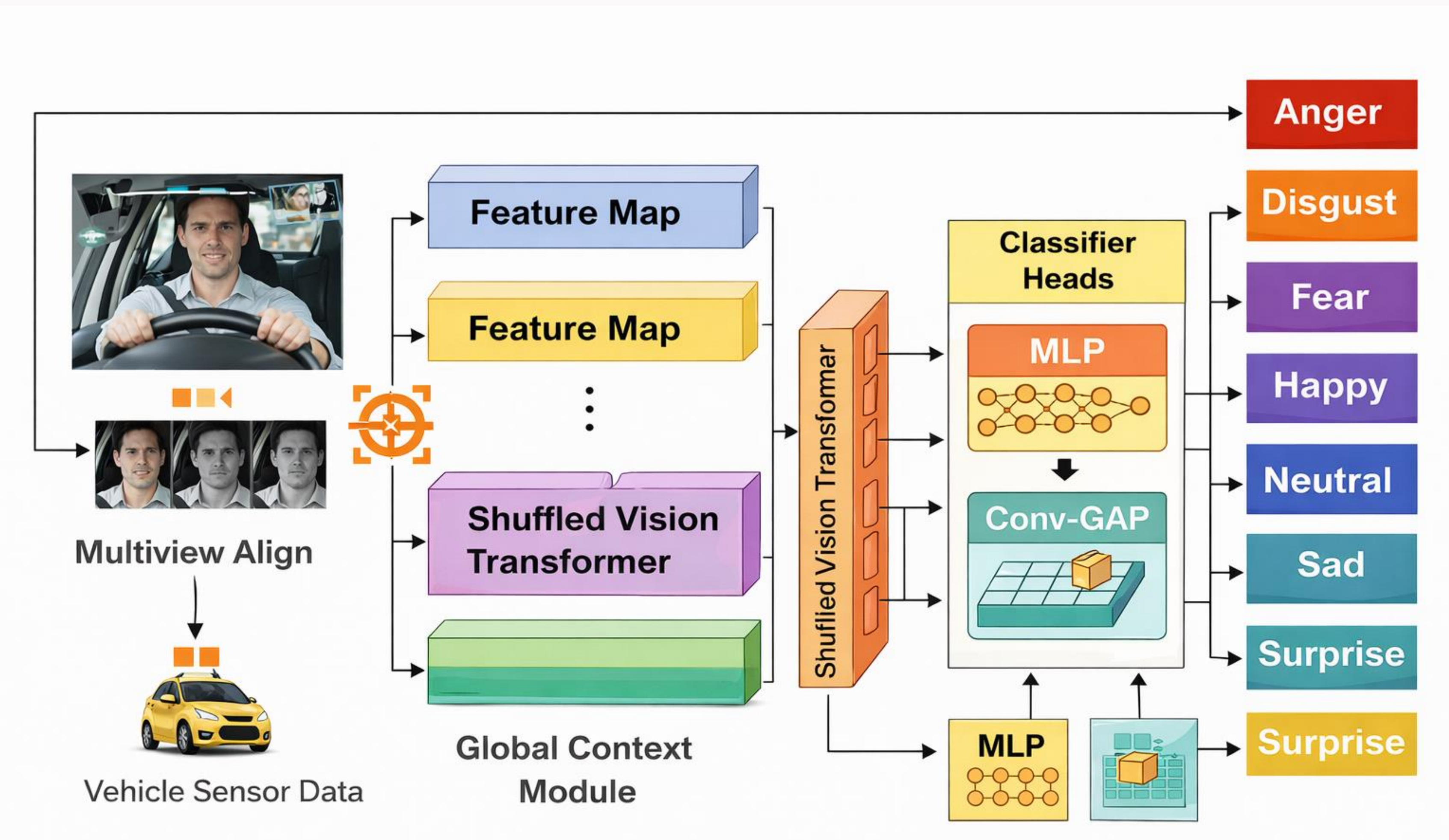


Fig. 2 Our proposed method ShuffViT-DFER Architecture with Extension v3

COMPARISON BETWEEN THE PROPOSED METHOD AND STATE OF THE ART AND OTHER EXISTING METHODS ON KMU-FED.

Methods	Accuracy
MPCNN 2018	86.90%
SqueezeNet 2022	86.86%
RBFNN 2022	88.80%
Efficient-swishNet DCNN 2022	85.50%
EfficientNet_FER 2023	88.17%
ShuffVit-(DFER) (Our Method MLP)	90.303%
ShuffVit-(DFER) (Our Method ConvGap)	91.212%

The proposed method outperformed all other methods on the KMU dataset.

4. Experiment Analysis and results

Our approach showed a comparable results and consistently outperforming all other approaches across different data splits on the KMU-FED dataset.

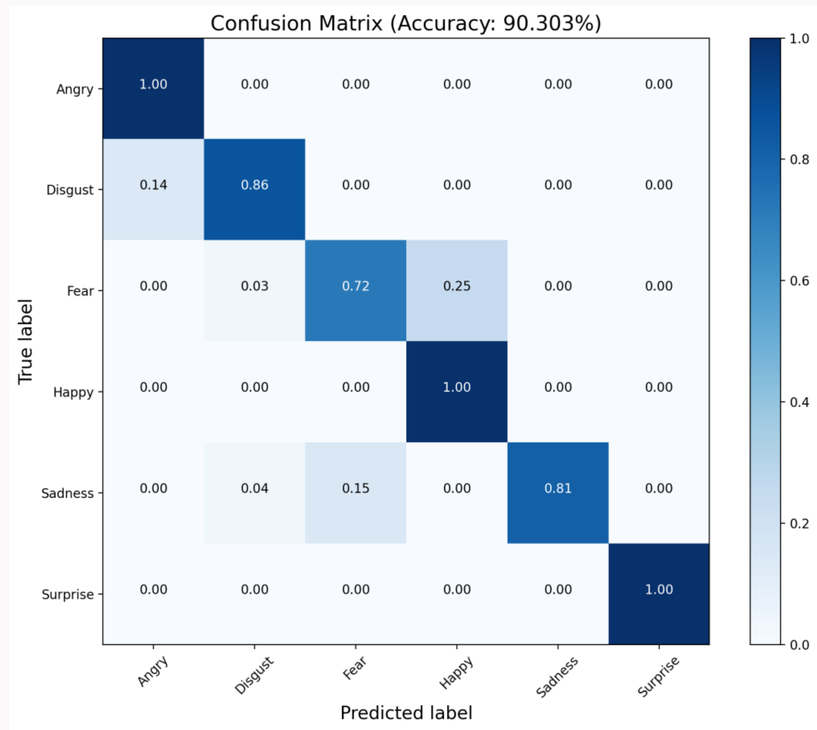


Fig. 3 Confusion Matrix with MLP

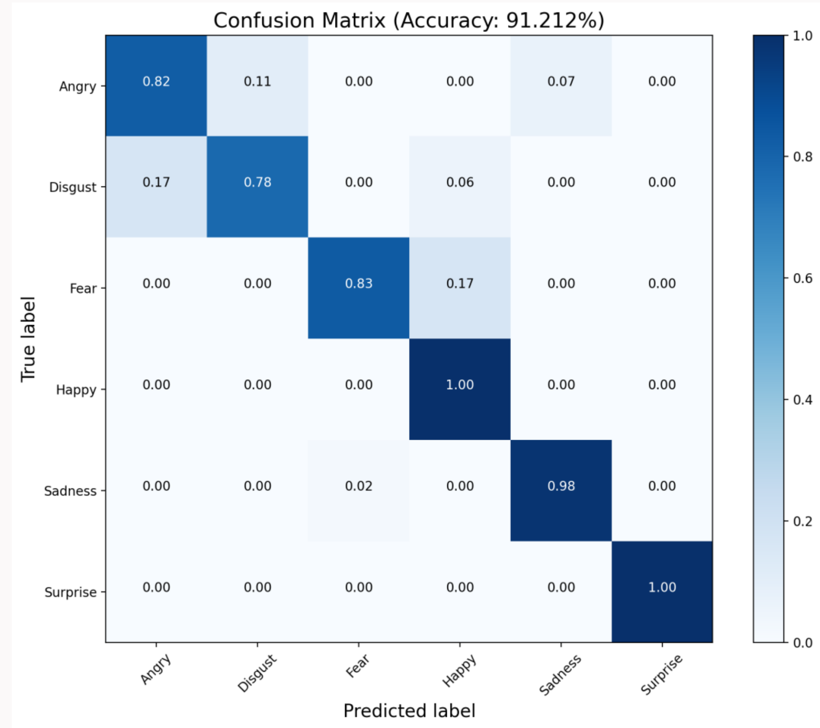


Fig. 4 Confusion Matrix with ConvGap

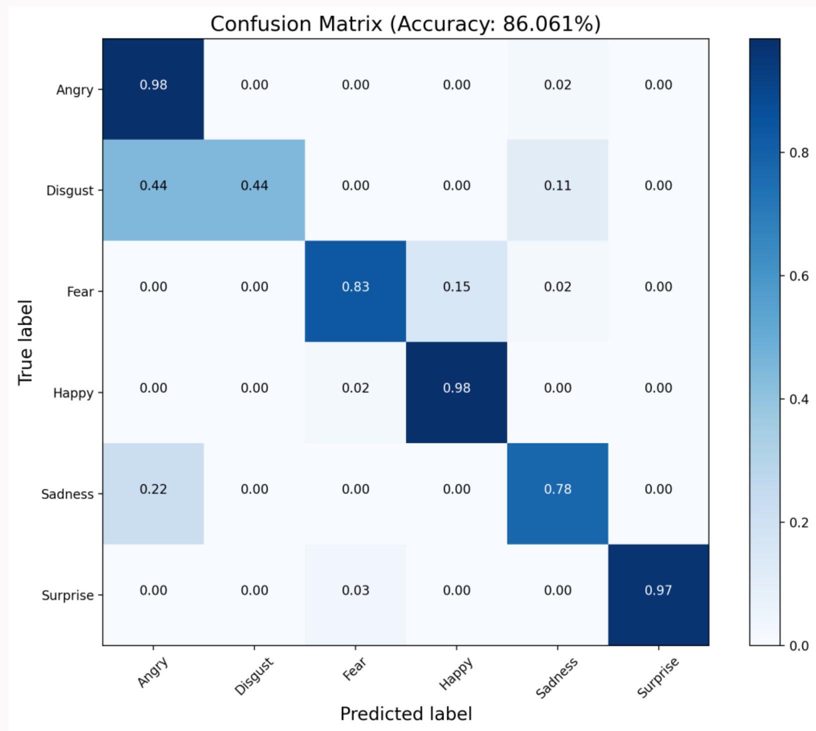
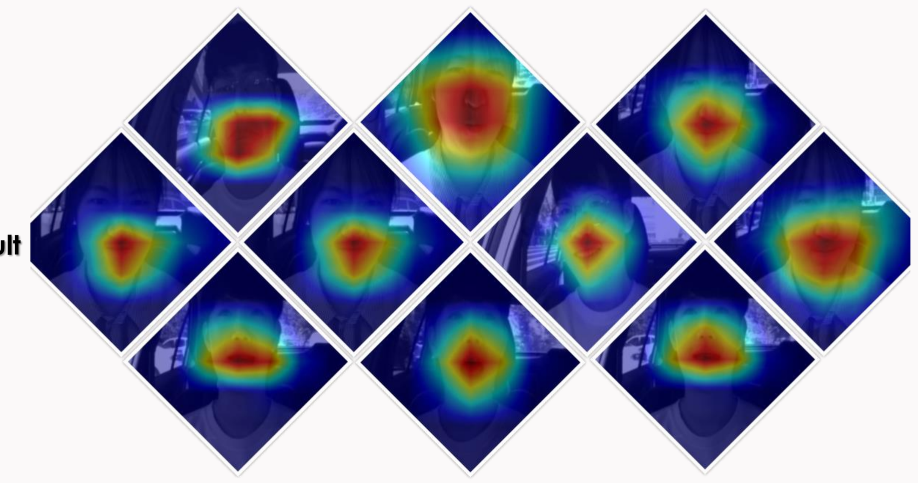


Fig. 5 Confusion Matrix with Baseline

Methods	Params (M)	Processing Time (ms)	Accuracy (%)	Precision	Recall	F1-score
ShuffleNet V2 w/o C	2.3	1.03	93.455	0.94	0.93	0.93
ShuffleNet V2 w C	2.0	1.11	90.455	0.90	0.91	0.91
EfficientViT -M2 w/o C	4.2	2.80	86.273	0.86	0.86	0.86
EfficientViT -M2 w C	4.7	2.70	95.727	0.96	0.96	0.96
ShuffVit-(DFER) (Our Method Baseline)	5.1	3.10	86.061	0.86	0.86	0.86
ShuffVit-(DFER) (Our Method MLP)	5.9	3.30	90.303	0.90	0.90	0.90
ShuffVit-(DFER) (Our Method ConvGap)	4.9	2.80	91.212	0.91	0.91	0.91

The proposed method outperformed all other methods on the KMU-FED dataset.

Fig. 6 Gradcam Result



5. Future Work

Several directions can be explored to further extend this work:

Multi-modal Fusion
Incorporating additional modalities such as **audio cues**, **vehicle sensor data**, or **physiological signals** could improve robustness, especially under challenging lighting and occlusion conditions.

Multi-view Facial Analysis
Extending the framework to handle **multi-camera inputs** can better address variations in head pose and view angles commonly encountered in real driving environments.

Cross-dataset Evaluation
Validating the proposed approach on additional DFER datasets would help assess generalization capability and domain robustness.

Temporal Modeling
Integrating temporal information using recurrent or attention-based sequence modeling could capture emotion dynamics over time, further improving recognition accuracy.

Edge Deployment Optimization
Future work will focus on deploying the model on **embedded and edge devices**, including model compression, quantization, and real-time performance evaluation in actual vehicle systems.

6. Conclusion

- We introduced ShuffViT-DFER, an efficient and fast method for recognizing driver's facial expressions.
- The obtained results demonstrated the effectiveness of the proposed approach
- Overall, the results confirm that **classifier head redesign plays a crucial role** in enhancing the effectiveness of hybrid CNN–ViT architectures, making the proposed ShuffViT-DFER extension a **practical and efficient solution** for real-time driver monitoring and intelligent safety systems.

7. References

[1] **Shuffle Vision Transformer: Lightweight, Fast and Efficient Recognition of Driver Facial Expression (Ibtisaam Saadi, Douglas W.Cunningham)**

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